

Julian Barberis

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ABOUT ME

Associate Degree graduate in Computer Programming from UNSAM and Academic Tutor for Algorithms 3. Strong foundations in algorithms, database design, and Clean Architecture. Specialized in building scalable end-to-end solutions: from robust backend APIs (Kotlin/Java with Spring Boot, TypeScript with NestJS) to modern frontend interfaces (React, Next.js) and AI-driven workflow automations.

PROJECTS

StudyQuest

2026

- Full-stack collaborative study and gamification platform built with **NestJS, TypeScript, React 19, PostgreSQL, and Docker**.
- Real-time matchmaking and party rooms via **WebSockets (Socket.IO)** for multi-user study session sync, rich chat, and live competitive quizzes.
- AI quest generation pipeline leveraging **Google Gemini 2.5 Flash** and an extraction sidecar to convert study notes and PDFs into structured trivia.
- Competitive **ELO** rating engine, dynamic leagues, skill trees, and **ACID** transactions in PostgreSQL for XP and leaderboard consistency.

BookLibre

2026

- Book-sharing platform built with **Kotlin, Spring Boot, React, and PostgreSQL**, structured around **Domain-Driven Design (DDD)** and Clean Architecture.
- 'BiblioKarma' gamified reputation engine with dynamic scoring based on polymorphic book categories, page count, and time-based reservation rules.
- Persistence layer with native **PostgreSQL** functions, views, and triggers, with schema evolution managed via **Flyway** migrations.
- Stateless **JWT with RBAC** (Role-Based Access Control) authentication, paired with Spring @Scheduled tasks for automated reservation lifecycles.

SQLify

2025

- Full-stack application using **React, TypeScript, Node.js, Express, and MySQL** to translate natural language questions into executable SQL queries.
- **Google Gemini AI** integration with dynamic schema context injection supporting complex relational database models.
- Secure execution sandbox with query normalization and sanitization, read-only guardrails, and real-time execution latency metrics.

AlgoQuePedir

2025

- Order management and logistical dispatch system in **Kotlin**, focused on advanced object-oriented design and algorithms.
- Complex data structures (**weighted graphs, priority heaps, and balanced trees**) for efficient delivery routing and order dispatching.
- Algorithmic paradigm implementation (**greedy heuristics, dynamic programming, and backtracking**), verified with a high-coverage unit test suite.

EXPERIENCE

Academic Tutor / Teaching Assistant — Algorithms 3

Argentina

UNSAM (Universidad Nacional de San Martín)

Aug 2026 - Present

- Academic tutoring and instruction in advanced data structures (balanced trees, heaps, graphs), asymptotic complexity, and algorithmic paradigms (Dynamic Programming, Greedy, Backtracking).
- Code reviews and technical mentorship on clean code standards, modularization, and object-oriented design patterns for student coursework.

- Technical blueprint drafting and engineering documentation for 50+ residential and commercial infrastructure projects with 100% regulatory compliance.
- Integration of AI-assisted workflows and automations for material takeoff calculations, schematic standardization, and repetitive task optimization in AutoCAD.

SKILLS

Languages & Backend: Kotlin, Java, TypeScript, JavaScript, Spring Boot, NestJS, Node.js, Python, C/C++
Frontend: React (19), Next.js, TypeScript, Tailwind CSS, Svelte, Zustand, Framer Motion, HTML5, CSS3
Databases & DevOps: PostgreSQL, MySQL, MongoDB, Redis, Flyway, TypeORM, Docker Compose, Git, Postman, Linux
AI & Architecture: LLM Integration (Gemini AI API, Claude), Clean Architecture, Domain-Driven Design (DDD), WebSockets

EDUCATION

UNSAM (Universidad Nacional de San Martín), Argentina
Associate Degree in Computer Programming

Mar 2023 - Jul 2026

Coderhouse, Argentina
Full Stack Developer Certification

Graduated 2023